

Maximum Likelihood Estimation for Multiple Linear Regression

COM3031 Probabilistic Machine Learning

1 Model Specification

We observe a dataset

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^n,$$

where $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector and $y_i \in \mathbb{R}$ is the response variable. We assume a linear model with Gaussian noise:

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Equivalently,

$$y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma^2 \sim \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2).$$

2 Likelihood Function

The probability density of a single observation is

$$p(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mathbf{w}^\top \mathbf{x}_i)^2\right).$$

Assuming independence across observations, the likelihood of the full dataset is

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{i=1}^n p(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma^2).$$

3 Log-Likelihood

Taking logarithms, the log-likelihood is

$$\ell(\mathbf{w}, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2.$$

The second term is the *sum of squared errors*.

4 MLE for the Weight Vector $\mathbf{w}$

Maximising the log-likelihood with respect to $\mathbf{w}$ is equivalent to minimising

$$\sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2.$$

Define:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the *design matrix*. Then

$$\sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Expanding:

$$(\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w}.$$

Taking the gradient with respect to $\mathbf{w}$:

$$\nabla_{\mathbf{w}} = 2\mathbf{X}^\top \mathbf{X} \mathbf{w} - 2\mathbf{X}^\top \mathbf{y}.$$

Setting the gradient to zero:

$$\mathbf{X}^\top \mathbf{X} \mathbf{w} = \mathbf{X}^\top \mathbf{y}.$$

These are the *normal equations*. Provided $\mathbf{X}^\top \mathbf{X}$ is invertible, the MLE is

$$\boxed{\mathbf{w}_{\text{MLE}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}$$

5 MLE for the Noise Variance σ^2

Differentiating the log-likelihood with respect to σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Setting the derivative to zero and substituting $\mathbf{w}_{\text{MLE}}$:

$$\boxed{\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} (\mathbf{X}\mathbf{w}_{\text{MLE}} - \mathbf{y})^\top (\mathbf{X}\mathbf{w}_{\text{MLE}} - \mathbf{y})}$$